

# CDFD Runtime Paper 07: Autonomous Discovery, Hypothesis Triage, and Falsification

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

May 2026

## Abstract

**Executive synthesis.** The discovery layer is a hypothesis factory, not an oracle. It perturbs runtime states, extracts features, proposes candidate regimes, and records falsification targets. This paper aligns the discovery manuscripts with the actual local experiment surface in `cdfd.runtime/experiments`: OOL phase mapping, finance-climate coupling, schema invention, vacuum hysteresis, knot stability, adaptive-surface instability, frontier sweeps, and gigamarathon coverage.

### Runtime evidence path

Prior CDFD mathematics  $\longrightarrow$  CDFL/runtime state  $\longrightarrow$  bounded output  $\longrightarrow$  falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

## 1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

## 2 Discovery Discipline

Autonomous discovery is useful only when it remains falsifiable. The CDFD The Runtime discovery layer does not turn a pattern into a law merely because the engine generated it. A discovery candidate needs:

- a script or CDFL model;
- saved output;
- parameter values;
- a pattern label;
- a falsification condition;
- a rerun command.

### 3 Discovery Modules

The discovery package contains feature extraction, hypothesis, generator, knowledge, sensitivity, detector, validator, and causal-discovery modules. The workflow is:

1. generate or perturb a state;
2. run bounded dynamics;
3. extract features such as mean  $\Psi_s$ , variance, collapse count, and stability index;
4. assign a candidate pattern label;
5. write a report or output file;
6. attempt falsification by changing parameters or null-coupling.

## 4 Selected Local Discoveries

### 4.1 Discovery A: OOL phase boundary

`experiments/notebooks/discovery_ool_phase.py` sweeps a two-dimensional prebiotic parameter space. The selected output `experiments/outputs/discovery_ool_phase_results.h5` contains grid points near  $\Lambda = 1$ . This is useful as a cross-Part-II phase-boundary bridge and as a test target for the Life Number. It is not empirical proof of abiogenesis.

### 4.2 Discovery B: finance-climate coupling

`experiments/notebooks/discovery_finance_climate.py` models a coupled finance-climate toy system and then runs causal discovery over the time series. The report frames the edge from financial  $\Psi$  volatility to climate constraint as a candidate causal result. The necessary null test is explicit: set coupling to zero and verify that the edge disappears.

### 4.3 Discovery C: schema invention

The schema-invention demonstration tests whether the knowledge layer can assign a new candidate label to a high- $\Psi_s$  anomaly. The narrow result is software capability: novelty detection and candidate schema recording. It is not, by itself, proof of a new physical regime.

## 4.4 Discovery D: vacuum hysteresis

`experiments/notebooks/discovery_vacuum_hysteresis.py` tests whether a high-flux pulse leaves localized  $M_s$  memory. The selected output records a center  $M_s$  peak of 15.5843. The proper status is candidate simulation result and measurement-protocol proposal.

## 4.5 Discovery E: n=7 knot stability

`experiments/notebooks/discovery_knot_symmetry.py` embeds a T(2,7) constraint pattern and tracks a stability index toward the CODATA 2022  $\chi^* = 137.035999177$  target. The release-selection note says to use it as a weak higher-knot self-stabilization stress test, not a proof.

## 4.6 Discovery F: adaptive-surface instability

`experiments/notebooks/discovery_adaptive_surface.py` produces an overloaded run. Because  $\Psi_s$  grows from approximately 5.15 to approximately 979 in the selected output, this is presented as an instability boundary and a safety test, not as an application success.

# 5 Frontier and Gigamarathon Sweeps

`experiments/outputs/frontier_sweep.h5` summarizes 2,500 parameter points and 2,487 non-trivial candidate records. This is hypothesis triage: which regions deserve closer study. `gigamarathon_1000_results.h5` covers 10 domains by 100 points each. This is coverage evidence for the discovery engine, not physical validation.

# 6 Promotion Ladder

A candidate discovery moves through this ladder:

1. simulation anomaly;
2. reproducible runtime diagnostic;
3. parameter-sensitivity report;
4. falsification attempt;
5. independent dataset or physical experiment;
6. domain-specific validation.

# 7 Conclusion

The discovery layer is valuable because it can search wider than a human can manually search. Its danger is overstatement. The runtime papers present discoveries as candidate simulation results with falsification handles until external validation exists.

## References

- [1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant*. CDFD Part I Paper I, preprint, 2026.
- [2] Steve Bico Mujjabi, MD. *The Universal Torus Knot Hierarchy:  $Z_n$  Symmetry, the Cinquefoil Family, and the General Structure of CDFT Particle Families*. CDFD Part I Paper VI, preprint, 2026.
- [3] Steve Bico Mujjabi, MD. *Even- $n$  Torus Knots in CDFT: The Exclusion of  $T(2, 2)$ , the Extension to  $T(2, 2m)$ , and the Anti-Phase Mass Scaling Law*. CDFD Part I Paper IX, preprint, 2026.
- [4] Steve Bico Mujjabi, MD. *The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law  $\Psi = \Phi/C$* . CDFD Part I Paper X, preprint, 2026.
- [5] Steve Bico Mujjabi, MD. *Friction, Superconductivity, Plasma States, and Transport Mysteries: Observable Tests of Adaptive Constraint*. CDFD Part I Paper XII, preprint, 2026.
- [6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics*. Public release archive, 2026, <https://doi.org/10.5281/zenodo.20250821>.